Unit-1

DC & AC Circuits

DC circuits: Ohm's and Kirchhoff's laws, analysis of series, parallel and series-parallel circuits excited by independent voltage sources for R, L, C parameters, current division, voltage division.

AC circuits: Generation of sinusoidal voltage, frequency of generated voltage, average value, RMS value, form, and peak factors. Real power, reactive power, apparent power, and Power factor.

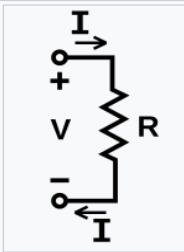
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1. Define ohms law and write its limitations

Statement

Ohm's law states that, at constant temperature, the current flowing through a conductor is directly proportional to the voltage across it.

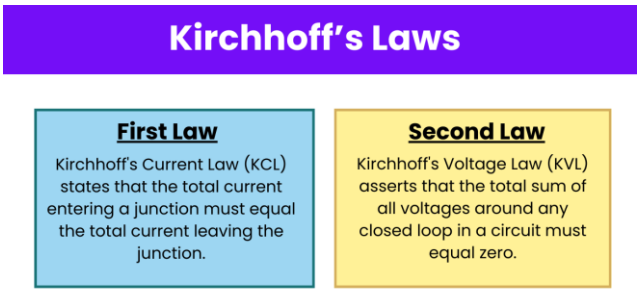
$$I \propto V \; ; I = \frac{V}{R} \; or \; V = IR$$



Limitations of Ohm's law

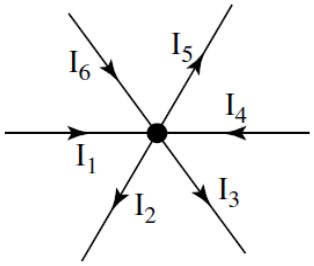
- Ohm's law is applicable when the temperature of the conductor is constant. Resistivity changes with temperature.
- The relation between voltage and current depends on the sign of voltage.
- It does not apply to semiconductors, which do not have a direct current-voltage relationship.
- Ohm's law is not applicable for unilateral electrical elements like diodes and transistors as they allow the current to flow through in one direction only.
- For non-linear electrical elements with parameters like capacitance, resistance etc the ratio of voltage and current won't be constant with respect to time making it difficult to use Ohm's law.

2. Explain Kirchoff's laws with examples



i. Kirchoff Current Law:

This law is applied at any node of an electric network. This law states that the algebraic sum of currents meeting at a junction or a node in a circuit is zero.

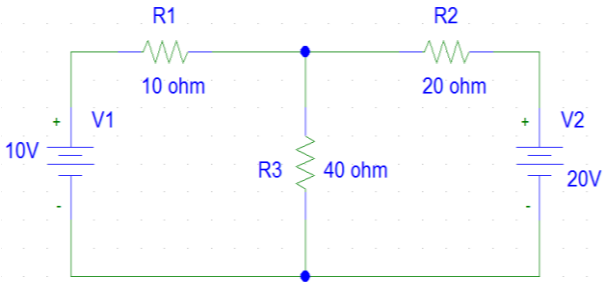


$$I_1 + I_6 + I_4 - I_5 - I_3 - I_2 = 0$$

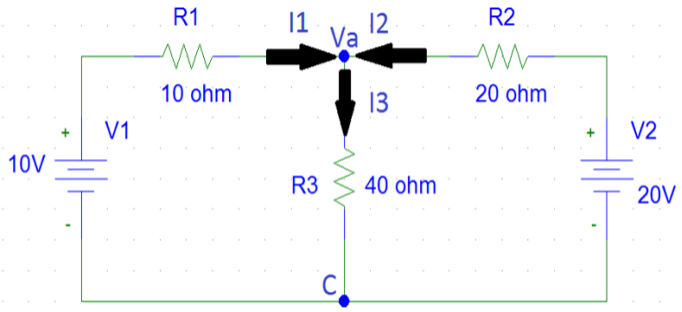
$$I_1 + I_6 + I_4 = I_5 + I_3 + I_2$$

The sum of current flowing towards a junction or a node is equal to the sum of currents flowing out of the junction. The current entering the junction has been taken as positive while the currents leaving the junction have been taken as negative.

Example:



Solution:



KCL equation:

Number of Nodes = Number of equations

1 Node= 1 Equation

The arrow marks i.e. Current directions can be decided randomly. Based on the arrow marks, we can write the current equation,

For example, I_1 is the incoming current to Node V_a,

$I_1 = \frac{Potential\ difference\ between\ two\ points}{Resistance} = \frac{V}{R} = \frac{V_1 - V_2}{R} = \frac{10 - V_a}{10}$; vice versa for I_2 & I_3

=

At Node Va,

Incoming Current = Outgoing current

$$I_1 + I_2 = I_3$$

$$\frac{(10-Va)}{10} + \frac{(20-Va)}{20} = \frac{Va}{40}$$

$$1 - \frac{Va}{10} + 1 - \frac{Va}{20} = \frac{Va}{40}$$

$$\frac{Va}{40} + \frac{Va}{10} + \frac{Va}{20} = 2$$

$$Va\left(\frac{1}{40} + \frac{1}{10} + \frac{1}{20}\right) = 2$$

$$Va\left(\frac{7}{40}\right) = 2$$

$$Va = 11.428V$$

$$I_1 = \frac{(10 - 11.428)}{10} = -0.143A$$

$$I_2 = \frac{(20 - 11.428)}{20} = 0.429A$$

$$I_3 = \frac{11.428}{40} = 0.286V$$

$I_1 + I_2 = I_3$

-0.143 + 0.429 = 0.286

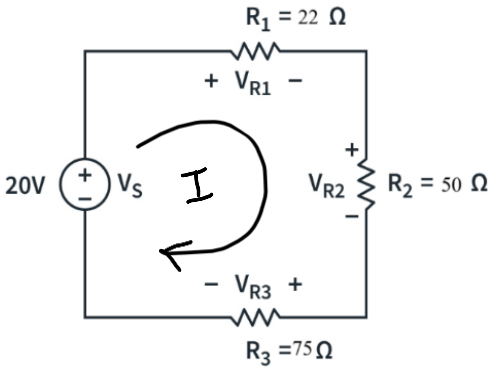
LHS =RHS

Kirchoff Current Law is proved

ii. Kirchhoff's voltage law:

This law is applicable to any closed loop in a circuit. KVL states that at any instant of time the algebraic sum of voltages in a closed loop is zero.

Example:1



Solution:

Number of closed loops = Number of equations

01 closed loop = 01 equation.

KVL

Note: - + = voltage rise

+ - = Voltage drop

Sum of Voltage rises = sum of voltage drops

$20 = V_{R1} + V_{R2} + V_{R3}$

$20 = 22I + 50I + 75\ I$ Note : V= IR, $V_{R1} = I \times 22 = 22I$ vice versa for V_{R2} & V_{R3}

$20 - 147\ I = 0$

$$I = \frac{20}{147} = 0.1360\ A$$ **(Current I is same for all resistances R_1, R_2, R_3. Because all 3 resistances are in series)**

Voltage across R_1, $V_{R1} = I\ R_1 = 0.1360 \times 22 = 2.9931\ V$

Voltage across R_2, $V_{R2} = I\ R_2 = 0.1360 \times 50 = 6.8027\ V$

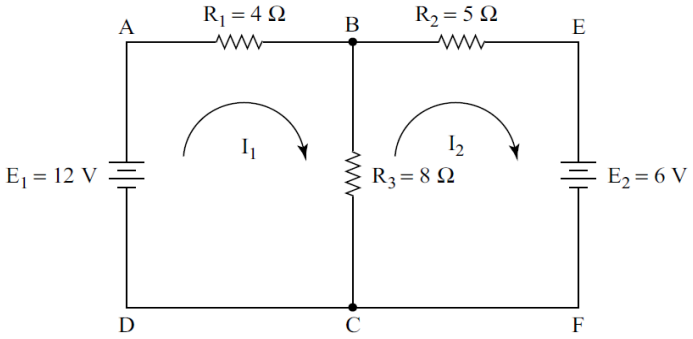
Voltage across R_3, $V_{R3} = I\ R_3 = 0.1360 \times 75 = 10.20\ V$

$V_s = V_{R1} + V_{R2} + V_{R3}$

$20 = 2.9931 + 6.8027 + 10.20$

LHS = RHS = Kirchoff Voltage law is verified

Example:2



Solution:

Number of closed loops = Number of equations

02 closed loop = 02 equation = 2 currents (I_1 & I_2)

Note: - + = voltage rise

+ - = Voltage drop

Apply KVL

Loop 1 (For loop DABCD, the voltage equation is)

Sum of Voltage rises = sum of voltage drops

$12 = V_{R1} + V_{R3}$

$12 = 4\ I_1 + 8\ (I_1 - I_2)$

$12 - 4\ I_1 - 8(I_1 - I_2) = 0$

$-12I_1 + 8I_2 = -12$ (1)

*Note : Current flowing through the resistor R_3 is the algebraic sum of the two currents I_1 and I_2. Here I_1 is flowing in the downward direction while I_2 is flowing in the upward direction. That's Why Voltage equation in R_3 is 8 (I_1-I_2).

Loop 2: For loop BEFCB, the voltage equation is

Sum of Voltage rises = sum of voltage drops

$$0 = V_{R2} + V_{R3} + 6$$

$$0 = 5\,I_2 + 8(I_2 - I_1) + 6$$

$$8I_1 - 13I_2 = -6 \qquad (2)$$

solving equations. (1) and (2), we get

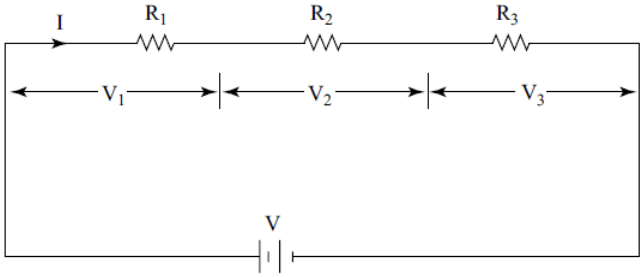
$$I_1 = 1.17\ A,\ I_2 = 0.26\ A$$

current flowing through R_3 is $(I_1 - I_2) = 0.91$ A

I_1 is flowing through R_1, I_2 is flowing through R_2 and $(I_1 - I_2)$ is flowing through R_3.

3. Problems on current division and voltage division

i) Voltage Division Rule



For easy calculation of voltage drop across resistors in a series circuit, a voltage divider rule is used.

$$I = \frac{V}{R_1 + R_2 + R_3}$$

$$V_1 = I\,R_1 = \frac{V}{R_1 + R_2 + R_3} \times R_1 = \frac{V}{R_T} \times R_1$$

$$R_T = R(\text{Total}) = R_1 + R_2 + R_3$$

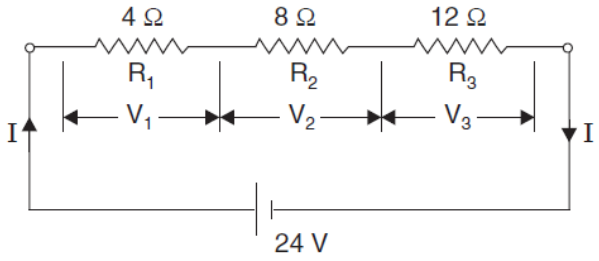
$$V_2 = I\,R_2 = \frac{V}{R_T} \times R_2$$

$$V_3 = I\,R_3 = \frac{V}{R_T} \times R_3$$

In general, the voltage across one in a series of resistors is given by

$$V_n = V_{total} \cdot \frac{R_n}{\Sigma R_i}$$

Example Problem:



Total resistance, $R = R1 + R2 + R3 = 4 + 8 + 12 = 24\ \Omega$

According to **voltage divider rule**, various voltage drops are:

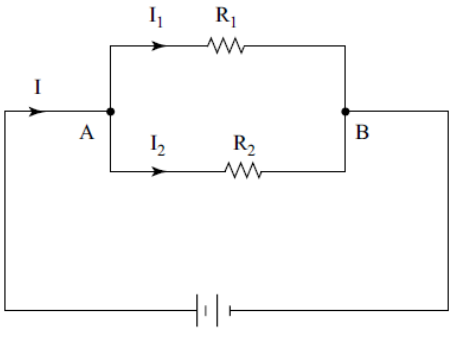
$$V_1 = V \times \frac{R_1}{R} = 24 \times \frac{4}{24} = 4\ \text{V}$$

$$V_2 = V \times \frac{R_2}{R} = 24 \times \frac{8}{24} = 8\ \text{V}$$

$$V_3 = V \times \frac{R_3}{R} = 24 \times \frac{12}{24} = 12\ \text{V}$$

ii) Current Division

Current divider rule is used in parallel circuits to find the branch currents if the total current is known.



Branch currents I_1, I_2

$$R_T = \frac{R_1 R_2}{R_1 + R_2} \qquad (1)$$

$$V = IR_T = I\frac{R_1 R_2}{R_1 + R_2} \qquad (2)$$

The current in each branch is given as

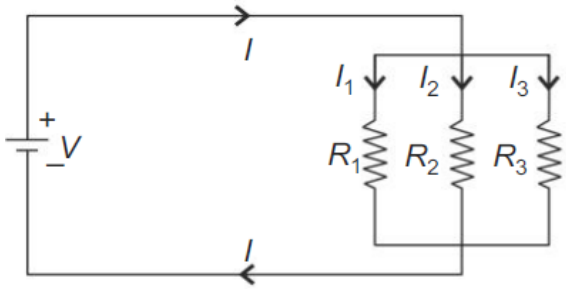
$$I_1 = \frac{V}{R_1} \; ; \quad I_2 = \frac{V}{R_2} \qquad (3)$$

Substitute equation (2) in I_1 & I_2 equations (3)

$$I_1 = \frac{V}{R_1} = I\frac{R_2}{R_1 + R_2}$$

$$I_2 = \frac{V}{R_2} = I\frac{R_1}{R_1 + R_2}$$

When Three Resistances are in Parallel,



$$I_1 = I \times \left(\frac{R_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1} \right)$$

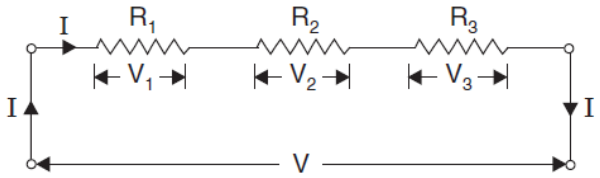
$$I_2 = I \times \left(\frac{R_3 R_1}{R_1 R_2 + R_2 R_3 + R_3 R_1} \right)$$

$$I_3 = I \times \left(\frac{R_1 R_2}{R_1 R_2 + R_2 R_3 + R_3 R_1} \right)$$

4. Problems on series – parallel combinations of R, L, C

Resistances in Series:

Current flowing through each resistance will be **same** but **voltage** drop across each of them will **vary** as per value of individual resistance.



Also, the sum of all the voltage drops ($V1 + V2 + V3$) is *equal to* the applied voltage (V).

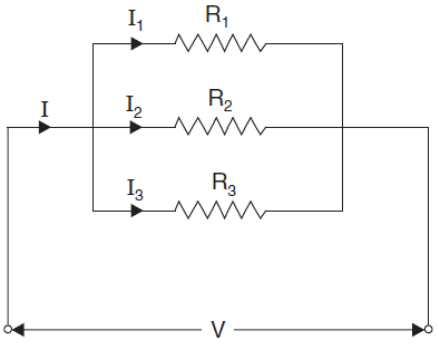
$$V = V_1 + V_2 + V_3$$

$$IR = IR_1 + IR_2 + IR_3$$

$$R = R_1 + R_2 + R_3$$

Resistances in Parallel

Voltage across each resistance will be **same** but **current** will be **different** depending upon the value of the individual resistance.



$$I = I_1 + I_2 + I_3$$

$$\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

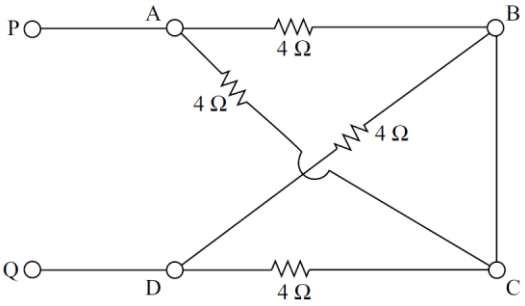
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

R is the *equivalent resistance* of the parallel combination.

$$R = \frac{R_1 R_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1}$$

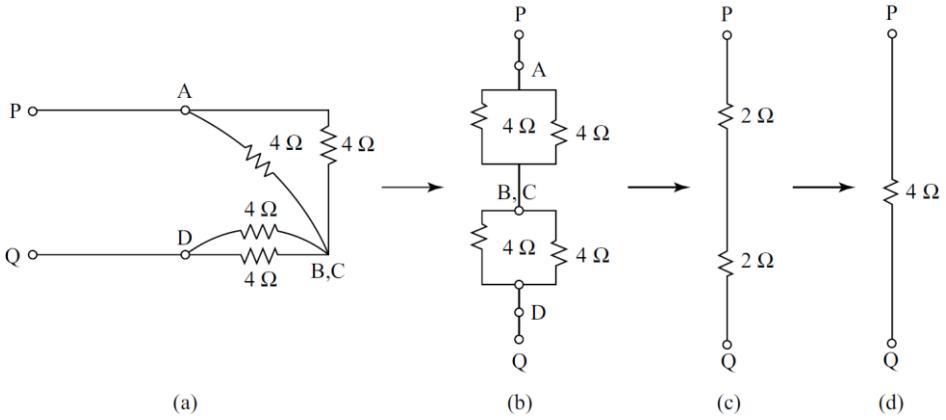
Example Problems:

Example 1. Calculate the resistance between the terminals P and Q of the network.



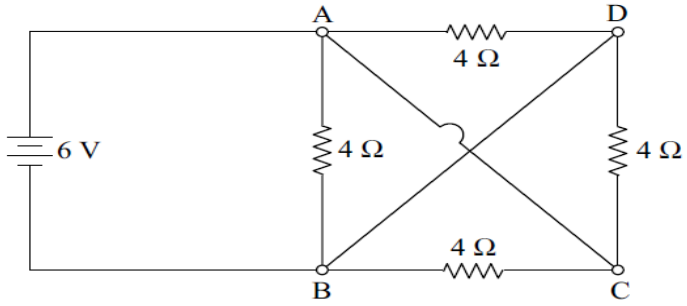
Solution:

There is no resistance in the branch B & C, so Let us bring points B and C together.



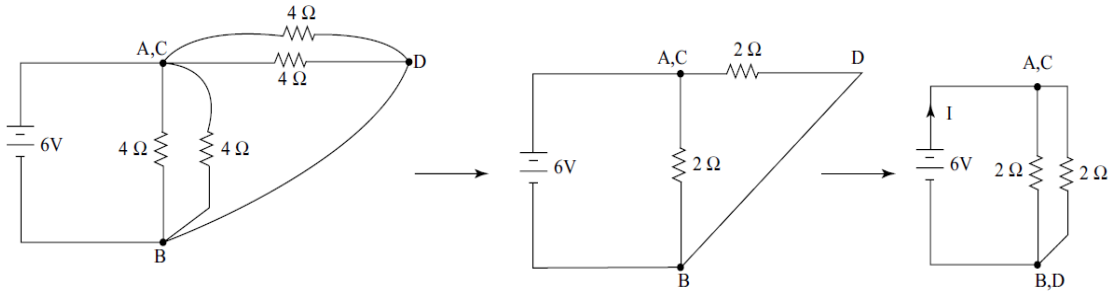
Answer = 4 ohm.

Example 2: Find the Req of the given circuit

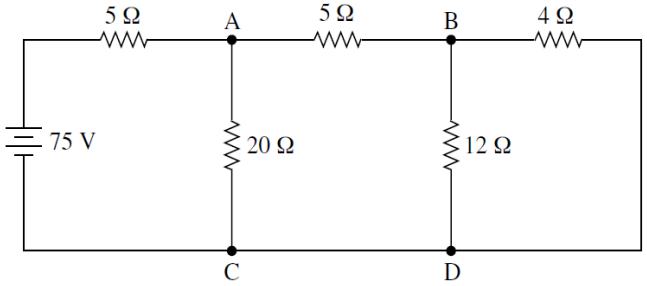


Solution:

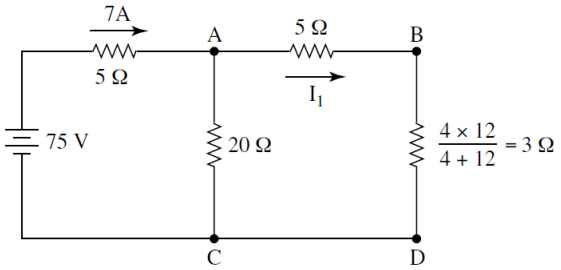
Points A and C are joined together. Similarly points B and D are joined together.



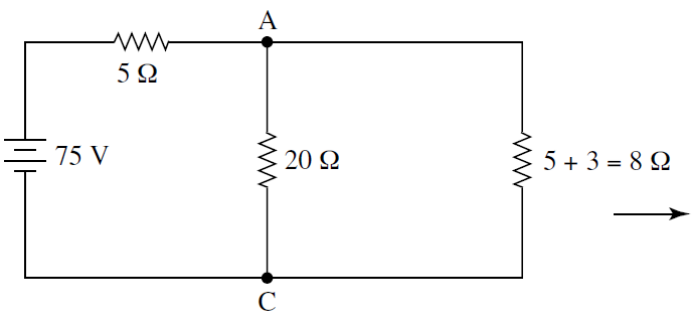
Example 3: Find the R_{eq} of the given circuit



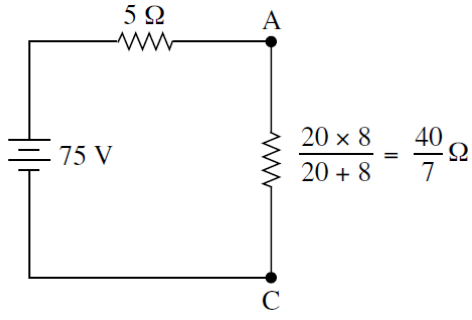
Solution:



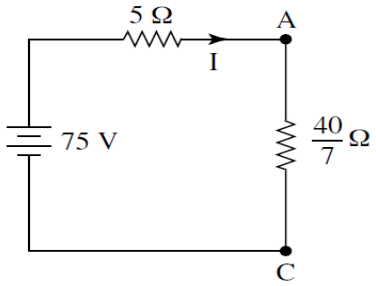
(a)



(b)



(c)

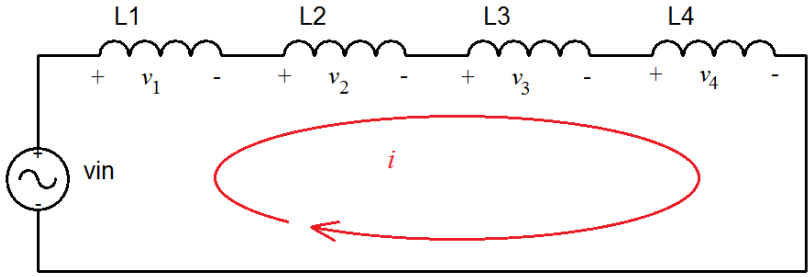


$$Req = 5 + \frac{40}{7} \rightarrow \quad \mathbf{R_{eq}} = \frac{75}{7}\ \Omega$$

(d)

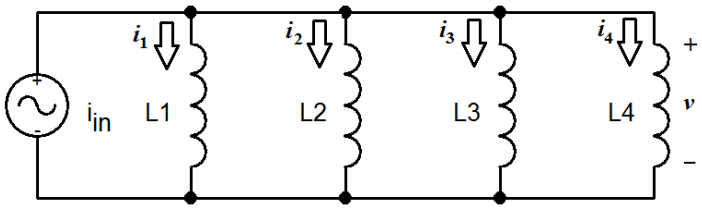
2. Inductors in Series

If inductors are connected in series (thus sharing the **same current**, then the total **voltage** dropped as the result of a change in current will be **additive** with each inductor.



When inductors are connected in series, the total inductance is the sum of the individual inductors' inductances.

$$\mathbf{L_{total} = L_1 + L_2 + \ldots L_n}$$

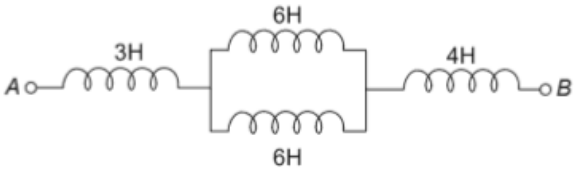


When inductors are connected in parallel, the total inductance is less than any one of the parallel inductors' inductances. he current through each parallel inductor will be a fraction of the total current, and the voltage across each parallel inductor will be equal.

The formula for calculating the parallel total inductance is the same form as for calculating parallel resistances.

$$\mathbf{L_{total}} = \frac{1}{\frac{1}{\mathbf{L_1}} + \frac{1}{\mathbf{L_2}} + \ldots \frac{1}{\mathbf{L_n}}}$$

Problem: Find the equvivalent L between A & B?



Solution:

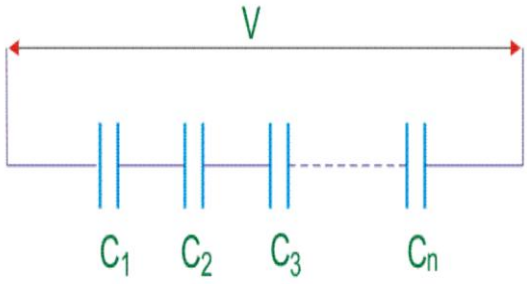
Step 1: 6 H & 6 H are in Parallel = $\frac{6 \times 6}{6+6} = 3$ **H**

Step 2: 3 H , 3 H , 4 H are in series = 3 + 3 + 4 = 10 H

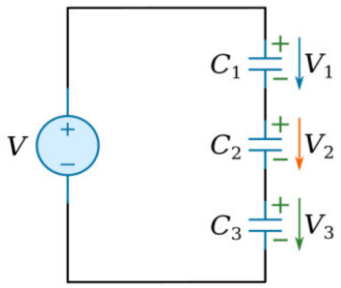
3. Capacitor in Series

In a series connection, capacitors decrease the total capacitance. As for any capacitor, the capacitance of the combination is related to charge and voltage.

$$C = \frac{Q}{V}.$$



(Or)



In series **Charge Q is same** but **Voltage drops are differed** with individual capacitances.

The sum of the capacitor voltages must equal the source voltage (Kirchhoff's voltage law)

$$V = V_1 + V_2 + V_3.$$

Entering the expressions for V_1, V_2, and V_3,

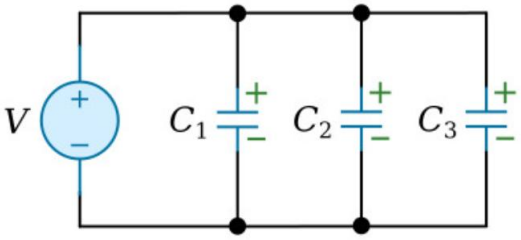
$$\frac{Q}{C_S} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}.$$

Total capacitance in series C_S is,

$$\frac{1}{C_S} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots,$$

Capacitors in Parallel

In parallel, the voltage across each capacitor is the same.



The total charge Q_t is the sum of the charges on each capacitor,

$$Q_t = Q_1 + Q_2 + Q_3$$

From the equation $C=Q/V$, it follows that $Q=CV$

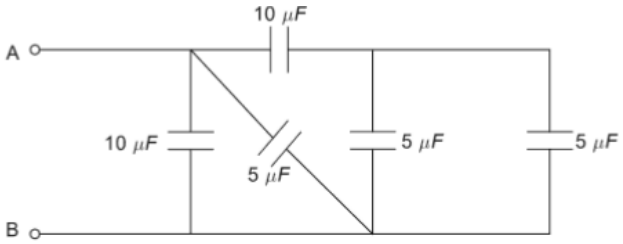
$$C_t V = C_1 V + C_2 V + C_3 V$$

In a parallel connection, capacitors increase the total capacitance, calculated by adding their individual capacitances.

$$C_t = C_1 + C_2 + C_3 + \ldots + C_n$$

Problem :

Find the equvivalent C between A & B ?



Solution:

Step 1: From the last end of circuit,

Capacitances 5 & 5 are in parallel = 5+5 = 10 μF

Step 2 : Then 10 & 10 become series $= \frac{10 \times 10}{10+10} = 5\ \mu F$

Step 3 : Then 5 & 5 become parallel, = 5+5 = 10 µF

Step 4 : Then 10 & 10 become parallel = 10 + 10 = 20 µF

The answer is C_t= 20 µF

For Remembrance:

Elements Symbol	RESISTOR	CAPACITOR	INDUCTOR
Denoted by	R	C	L
Equation	$R = \frac{V}{I}$	$C = \frac{Q}{V}$	$L = \frac{V_L}{(di/dt)}$
Series	$R_T = R_1 + R_2$	$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}$	$L_T = L_1 + L_2$
Parallel	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$	$C_T = C_1 + C_2$	$\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2}$

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5. Derive the equation for Rms value, average value, peak factor and form factor for sinusoidal waveform.

i) RMS Value:

Definition

The RMS value of AC is the equivalent DC value that produces the same heating effect in a circuit as the AC produces for the same time.

Derivation:

RMS current is defined as the square root of the mean of the squares of the instantaneous current over a cycle.

$$i = I_m \sin \theta$$

Square of the current $i = I_m \sin\theta$ is $I_m^2 \sin^2\theta$

Its mean over one cycle is calculated by integrating it from 0 to 2π and dividing by the time period of 2π.

RMS value, $I = \sqrt{1/2\pi \int_0^{2\pi} I_m^2 \sin^2\theta \, d\theta}$

To calculate, let us put $\sin^2\theta = \frac{1-\cos 2\theta}{2}$

then $I = \sqrt{\frac{I_m^2}{4\pi} \int_0^{2\pi} (1-\cos 2\theta)\, d\theta}$

$$= \sqrt{\frac{I_m^2}{4\pi}\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{2\pi}}$$

$$= \sqrt{\frac{I_m^2}{4\pi} \times 2\pi}$$

$$= \sqrt{\frac{I_m^2}{2}} = \frac{I_m}{\sqrt{2}}$$

Therefore, $I = \frac{I_m}{\sqrt{2}} = 0.707\, I_m$

$$\text{RMS value} = \frac{\text{Maximum value}}{\sqrt{2}} \quad \text{(for a sinusoidal wave)}$$

ii) Average value

Definition

The average value of an AC current is the value of a steady (DC) current that would transfer the same total amount of electric charge through a circuit in the same time as the AC current does.

Average value has to be calculated by considering one-half cycle as,

$$I_{av} = \frac{1}{\pi} \int_0^{\pi} I_m \sin\theta \, d\theta$$

$$= \frac{I_m}{\pi} \int_0^{\pi} I_m \sin\theta \, d\theta$$

$$= \frac{I_m}{\pi} [-\cos\theta]_0^{\pi}$$

$$= -\frac{I_m}{\pi} [-1 - 1]$$

$$= \frac{2 I_m}{\pi} = 0.637 I_m$$

Similarly average value for sinusoidal voltage, $V_{av} = 0.637 V_m$

Therefore, average value $\frac{2 I_m}{\pi}$ or $= \frac{2 V_m}{\pi}$

iii) Form factor

Form factor is an indicator of the shape or the form of the ac wave. It is the ratio of the RMS value to the average value of an alternating quantity. For a sinusoidal varying quantity, the form factor K_f is

$$K_f = \frac{\text{RMS value}}{\text{Average value}} = \frac{0.707 I_m}{0.637 I_m} = 1.11$$

iv) Peak Factor:

The peak or crest factor is the ratio of peak or maximum value to its rms value

$$\text{Peak factor} = \frac{\text{Peak value}}{\text{RMS value}}$$

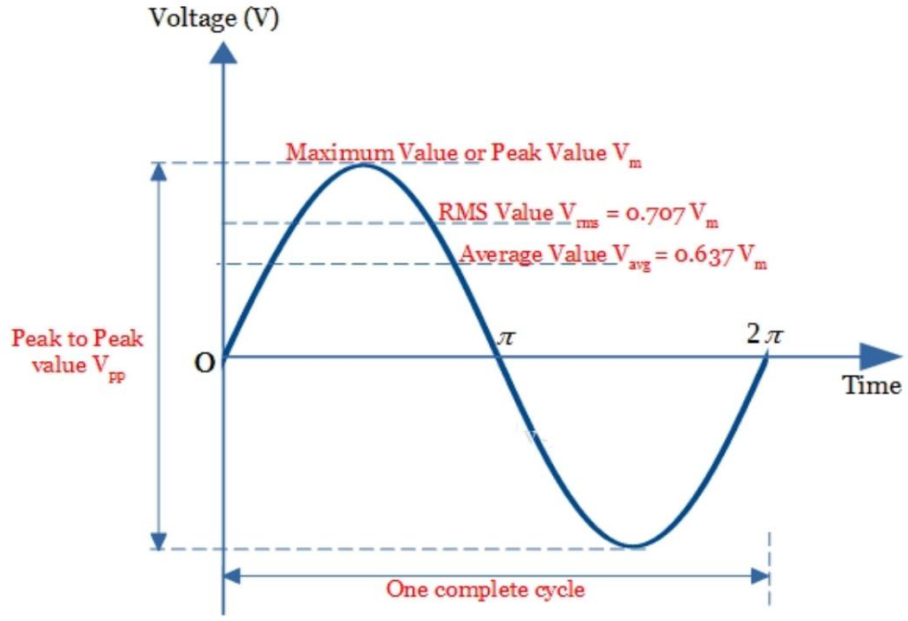
$$\text{Peak factor } (K_f) = \frac{I_m}{I_m/\sqrt{2}} = \sqrt{2} = 1.414.$$

6. Explain the following terms

a. Cycle b. Amplitude c. Time period. D. Frequency. E. Instantaneous value. F. Peak to peak value

a. Cycle.

One complete set of positive and negative values of an alternating quantity is known as a *cycle*. A cycle may also sometimes be specified in terms of angular measure. One complete cycle is said to spread over 360° or 2π radians.



b. Amplitude. The maximum value, positive or negative of an alternating quantity, is known as its *amplitude*.

c. Frequency (f). The number of cycles per second is called the frequency of the alternating quantity. Its unit is *hertz* (Hz).

d. Time Period (T). The time taken by an alternating quantity to complete one cycle is called its *time period*. It is measured in seconds.

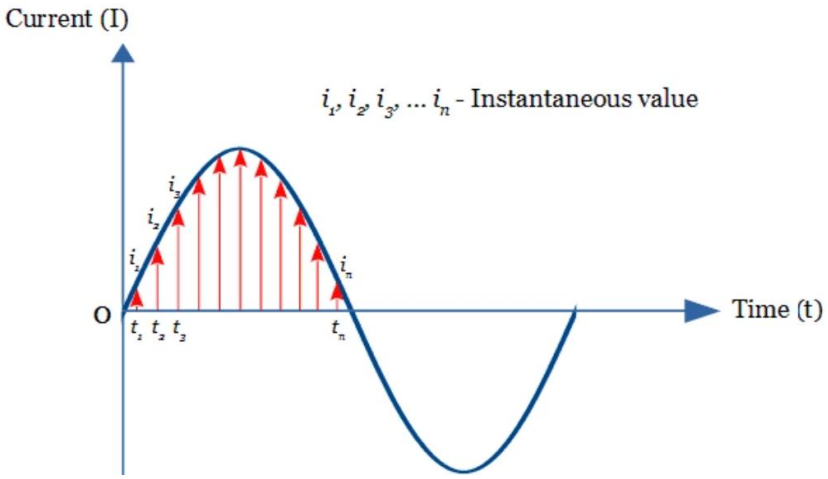
e. Instantaneous value.

The instantaneous value can be defined as the value of an alternating quantity at any particular instant. and is denoted for current as “i” and voltage as “e”.

The instantaneous value of a sinusoidal waveform is generalized by the formula,

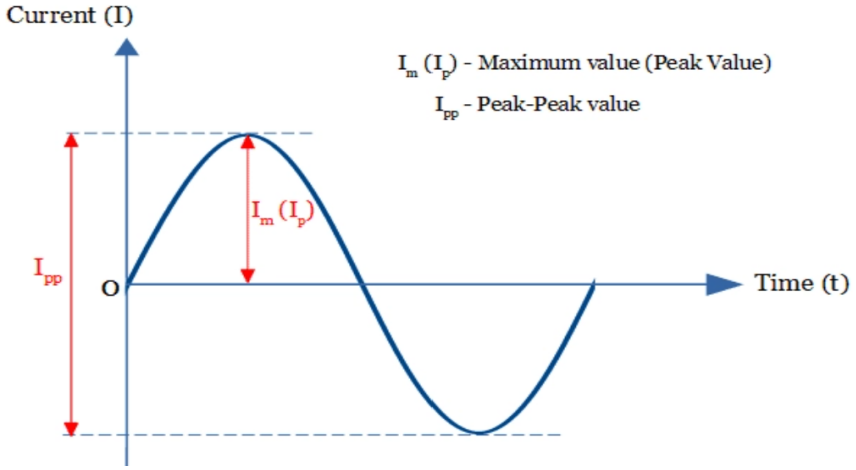
$$e = E_m sin\theta$$

$$i = I_m sin\theta$$



F. Peak to peak value

The sum of peak values of the positive and negative half cycle is known as peak-to-peak value. It is expressed as I_{pp}.



7. Explain the power triangle and importance of power factor

The **power triangle** graphically shows the relationship between real power(*P*), reactive power (*Q*) and apparent power (*S*).

Real (or) true power (or) active power (P)

It is the power which is dissipated in the circuit resistance. The useful power that actually does work (measured in watts, W).

Active power, $P = VI \cos\phi$ Watts

Reactive power (Q)

It is the power developed in the inductive reactance of the circuit. The power that flows back and forth due to inductors or capacitors, not doing useful work. (measured in VAR).

$Q = VI \sin\phi$ VAR

Apparent power (S)

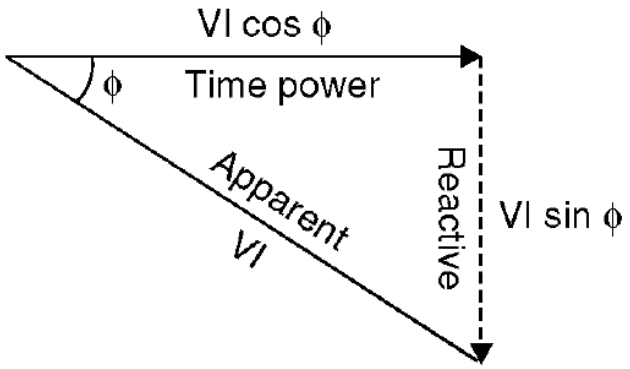
It is defined as the product of the RMS value of voltage (V) and current (I). It is denoted by S.

Apparent power, S = Voltage × Current

S = VI VA

Power Triangle:

These three powers are shown in the power triangle of Figure.



As per the right triangle formula,

$$S^2 = P^2 + Q^2$$

$$S = \sqrt{P^2 + Q^2}$$

$$S = \sqrt{(VI\cos\phi)^2 + (VI\sin\phi)^2}$$

$$S = \sqrt{V^2I^2\cos^2\phi + V^2I^2\sin^2\phi}$$

Factor out V^2I^2:

$$S = \sqrt{V^2I^2(\cos^2\phi + \sin^2\phi)}$$

Use the identity:

$$\cos^2\phi + \sin^2\phi = 1$$

$$S = \sqrt{V^2I^2(1)}$$

$$S = \sqrt{V^2I^2}$$

$$S = VI$$

Power factor:

It is ratio of **real or active power (P)** to the **apparent power (S)**.

$$\text{Power Factor} = \cos\phi = \frac{\text{Real Power (P)}}{\text{Apparent Power (S)}}$$

It is also defined as the cosine of the phase angle between the voltage and current. cosØ is known as power factor.

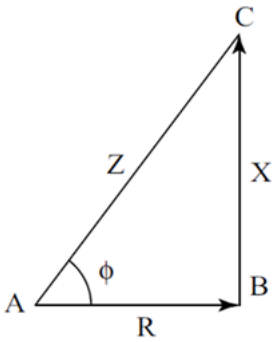
Importance of Power Factor:

1. Power factor is the measure of effective utilization of power.
2. It is important to have power factor as close to unity as possible.
3. A good power factor reduces power losses in cables and equipment.
4. A good power factor increases the overall efficiency of the power system.
5. Industries pay extra money if their power factor is low, so improving power factor reduces the electricity bill.

8. What is impedance and explain the impedance triangle

Impedance is the total opposition offered to the flow of alternating current in a circuit. It includes both resistance (R) and reactance (X).
Resistance (R) opposes current in both AC and DC circuits.
Reactance (X) opposes current only in AC circuits due to inductors and capacitors.

- If the circuit is **inductive**, $X = X_L$
- If the circuit is **capacitive**, $X = -X_C$(negative because capacitive reactance opposes inductive reactance)



Impedance Triangle

Since the impedance triangle is a right triangle:

$$Z^2 = R^2 + X^2$$

Taking square root on both sides:

$$Z = \sqrt{R^2 + X^2}$$

or, $Z = R + jX_L$

Where j indicates rotation by 90° in anti-clockwise direction.

$$\text{Power Factor} = \cos\phi = \frac{R}{Z}$$

In circuits with **inductive reactance (X_L) –** such as motors, transformers.

- Current lags the voltage.
- The power factor is lagging power factor.

In circuits with **capacitive reactance (X_C)**, such as capacitor banks:

- Current leads voltage.
- The power factor is leading power factor.
